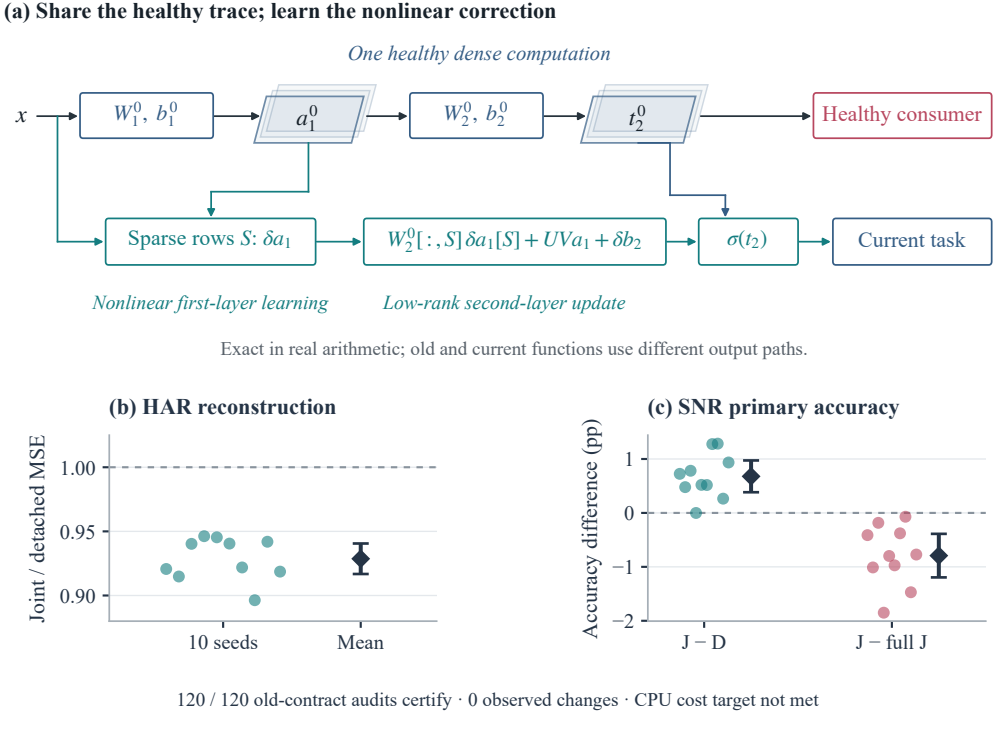

(a) Share the healthy trace; learn the nonlinear correction
One healthy dense computation
x
$W_1^0,\ b_1^0$
a_1^0
$W_2^0,\ b_2^0$
t_2^0
Healthy consumer
Sparse rows S: δa_1
$W_2^0[:, S]\,\delta a_1[S] + UVa_1 + \delta b_2$
$\sigma(t_2)$
Current task
Nonlinear first-layer learning
Low-rank second-layer update
Exact in real arithmetic; old and current functions use different output paths.
(b) HAR reconstruction
Joint / detached MSE
1.00
0.95
0.90
10 seeds
Mean
(c) SNR primary accuracy
Accuracy difference (pp)
1
0
−1
−2
J − D
J − full J
120 / 120 old-contract audits certify · 0 observed changes · CPU cost target not met